

COMPLETE MATHEMATICAL LECTURE NOTES ON
METRIC SPACES & MATRIX SPACES

Axioms, Semi-Metrics, Open Spheres, Topological Theorems, Limit Points
& Matrix Norms

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1 Module 1: Foundations of Metric Spaces and Semi-Metrics

A metric space is a mathematical abstraction that formalizes and generalizes the intuitive geometric concept of distance between points in any set.

1.1 Definition of a Metric Space

Definition: Metric and Metric Space

Let X be a non-empty set. A mapping $d : X \times X \rightarrow \mathbb{R}$ is called a **metric** (or distance function) on X if and only if it satisfies the following four fundamental axioms for all elements $x, y, z \in X$:

[M1] **Non-negativity:**

$$d(x, y) \geq 0, \quad \forall x, y \in X \quad (1)$$

[M2] **Identity of Indiscernibles (Definiteness):**

$$d(x, y) = 0 \iff x = y \quad (2)$$

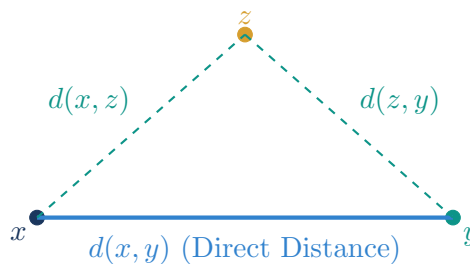
[M3] **Symmetry:**

$$d(x, y) = d(y, x), \quad \forall x, y \in X \quad (3)$$

[M4] **Triangle Inequality:**

$$d(x, y) \leq d(x, z) + d(z, y), \quad \forall x, y, z \in X \quad (4)$$

If d is a metric on X , then the ordered pair (X, d) is called a **Metric Space**, and the non-negative real number $d(x, y)$ is called the **distance between x and y** .



Geometric Axiom [M4]: $d(x, y) \leq d(x, z) + d(z, y)$

1.2 Semi-Metric (Pseudo-Metric)

Definition: Semi-Metric / Pseudo-Metric

A mapping $d : X \times X \rightarrow \mathbb{R}$ is called a **semi-metric** (or **pseudo-metric**) if and only if it satisfies:

$$[\text{M1}] \quad d(x, y) \geq 0, \quad \forall x, y \in X.$$

$$[\text{M2}'] \quad d(x, x) = 0, \quad \forall x \in X \quad (\text{Note: } d(x, y) = 0 \text{ does **not** require } x = y).$$

$$[\text{M3}] \quad d(x, y) = d(y, x), \quad \forall x, y \in X.$$

$$[\text{M4}] \quad d(x, y) \leq d(x, z) + d(z, y), \quad \forall x, y, z \in X.$$

Remark: Every metric is a semi-metric, but a semi-metric is generally **not** a metric because distinct points may have zero distance.

Example: A Semi-Metric that is NOT a Metric

Question: Consider the mapping $d : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ defined by:

$$d(x, y) = |x^2 - y^2|, \quad \forall x, y \in \mathbb{R} \quad (5)$$

Show that d is a semi-metric on $\mathbb{R}$, but is **not** a metric.

Verification of Axioms:

1. **[M1] Non-negativity:** Since absolute value is always non-negative, $d(x, y) = |x^2 - y^2| \geq 0, \forall x, y \in \mathbb{R}$.
2. **[M3] Symmetry:** $d(x, y) = |x^2 - y^2| = |-(y^2 - x^2)| = |y^2 - x^2| = d(y, x)$.
3. **[M4] Triangle Inequality:** For any $x, y, z \in \mathbb{R}$:

$$\begin{aligned} d(x, y) &= |x^2 - y^2| = |(x^2 - z^2) + (z^2 - y^2)| \\ &\leq |x^2 - z^2| + |z^2 - y^2| = d(x, z) + d(z, y) \end{aligned} \quad (6)$$

4. **[M2] Failure of Indiscernibles:**

$$\begin{aligned} d(x, y) = 0 &\iff |x^2 - y^2| = 0 \iff x^2 = y^2 \\ &\iff x = y \quad \text{or} \quad x = -y \end{aligned} \quad (7)$$

For instance, choose $x = 2$ and $y = -2$:

$$d(2, -2) = |2^2 - (-2)^2| = |4 - 4| = 0$$

However, $2 \neq -2$. Therefore, condition $[M2]$ fails.

Conclusion: $d(x, y) = |x^2 - y^2|$ is a **semi-metric** (pseudo-metric), but **not a metric** on $\mathbb{R}$. □

1.3 Standard Metrics on Euclidean Spaces

Standard Metrics on $\mathbb{R}^2$

Let $x = (x_1, y_1)$ and $y = (x_2, y_2)$ be points in $\mathbb{R}^2$.

1. **Euclidean Metric (d_2):**

$$d_2(x, y) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (8)$$

2. **Taxicab / Manhattan Metric (d_1):**

$$d_1(x, y) = |x_1 - x_2| + |y_1 - y_2| \quad (9)$$

3. **Maximum / Chebyshev Metric (d_∞):**

$$d_\infty(x, y) = \max(|x_1 - x_2|, |y_1 - y_2|) \quad (10)$$

1.4 Discrete Metric Space

Definition: Discrete Metric Space

Let X be any arbitrary non-empty set. The **discrete metric** on X is defined as:

$$d(x, y) = 0 \quad \text{if } x = y, \quad \text{and} \quad d(x, y) = 1 \quad \text{if } x \neq y \quad (11)$$

The pair (X, d) is called a **Discrete Metric Space**. In this space, every single point is isolated at a distance of 1 from every other point.

1.5 Matrix Spaces as Metric Spaces

Let $M_{m \times n}(\mathbb{R})$ denote the vector space of all $m \times n$ real matrices.

Definition: Frobenius Metric on Matrix Space

For any matrices $A, B \in M_{m \times n}(\mathbb{R})$, the **Frobenius distance** is defined by:

$$d_F(A, B) = \|A - B\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n (a_{ij} - b_{ij})^2} = \sqrt{\text{Tr}((A - B)^T(A - B))} \quad (12)$$

Properties:

- $d_F(A, B) \geq 0$ and $d_F(A, B) = 0 \iff A = B$.
- $d_F(A, B) = d_F(B, A)$.
- $d_F(A, B) \leq d_F(A, C) + d_F(C, B)$ (follows from Cauchy-Schwarz on matrix trace).
- $(M_{m \times n}(\mathbb{R}), d_F)$ is isometrically isomorphic to the Euclidean space $(\mathbb{R}^{mn}, d_2)$ and is therefore a **complete metric space**.

2 Module 2: Open Spheres, Neighborhoods, and Interior Points

2.1 Open Spheres (Open Balls)

Definition: Open Sphere / Open Ball

Let (X, d) be a metric space, $x_0 \in X$, and $r > 0$ be a positive real number. The **open sphere** (or open ball) centered at x_0 of radius r is the subset:

$$S_r(x_0) = B(x_0, r) = \{x \in X \mid d(x, x_0) < r\} \quad (13)$$

The **closed sphere** centered at x_0 of radius r is:

$$\bar{S}_r(x_0) = \bar{B}(x_0, r) = \{x \in X \mid d(x, x_0) \leq r\} \quad (14)$$

2.2 Geometric Shapes of Open Spheres in Different Metrics

Examples: Open Spheres with $x_0 = (0, 0)$ and $r = 1$

1. **Usual Metric on $\mathbb{R}$ with $x_0 = 0, r = 1$:**

$$S_1(0) = \{x \in \mathbb{R} \mid d(x, 0) < 1\} = \{x \in \mathbb{R} \mid |x| < 1\} = (-1, 1) \quad (\text{Open Interval})$$

2. **Euclidean Metric $(\mathbb{R}^2, d_2)$ with $x_0 = (0, 0), r = 1$:**

$$S_1(0, 0) = \{(x, y) \in \mathbb{R}^2 \mid \sqrt{x^2 + y^2} < 1\} = \{(x, y) \mid \mathbf{x}^2 + \mathbf{y}^2 < 1\} \quad (\text{Circular Disk})$$

3. **Taxicab Metric $(\mathbb{R}^2, d_1)$ with $x_0 = (0, 0), r = 1$:**

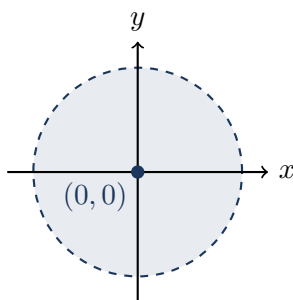
$$S_1(0, 0) = \{(x, y) \in \mathbb{R}^2 \mid |x| + |y| < 1\} \quad (\text{Rotated Square / Diamond})$$

4. **Maximum Metric $(\mathbb{R}^2, d_\infty)$ with $x_0 = (0, 0), r = 1$:**

$$S_1(0, 0) = \{(x, y) \in \mathbb{R}^2 \mid \max(|x|, |y|) < 1\} = (-1, 1) \times (-1, 1) \quad (\text{Axis-aligned Square})$$

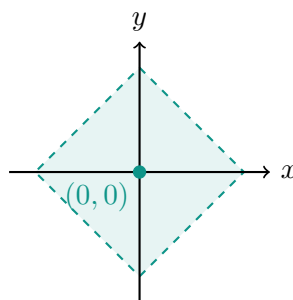
5. **Discrete Metric Space (X, d) :**

$$S_r(x_0) = \{x_0\} \quad \text{if } r \leq 1, \quad \text{and} \quad S_r(x_0) = X \quad \text{if } r > 1$$



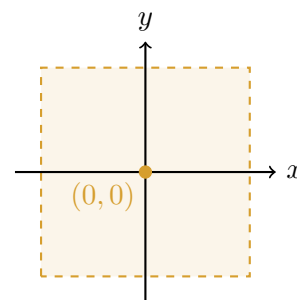
Euclidean Ball (d_2)

$$x^2 + y^2 < 1$$



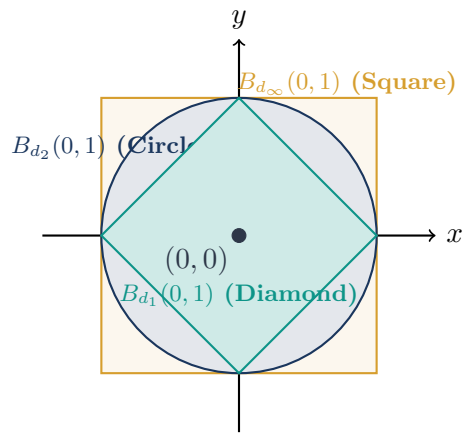
Taxicab Ball (d_1)

$$|x| + |y| < 1$$



Maximum Ball (d_∞)

$$\max(|x|, |y|) < 1$$



Inclusion Hierarchy: $B_{d_1}(0,1) \subset B_{d_2}(0,1) \subset B_{d_\infty}(0,1)$

2.3 Neighborhoods and Interior Points

Definitions: Neighborhood and Open Neighborhood

Let (X, d) be a metric space and $x \in X$.

- A subset $N \subseteq X$ is called a **neighborhood of x** if and only if there exists an open sphere $S_r(x)$ centered at x with radius $r > 0$ such that:

$$S_r(x) \subseteq N \quad (15)$$

- If the neighborhood N is itself an open set, it is called an **open neighborhood** of x .

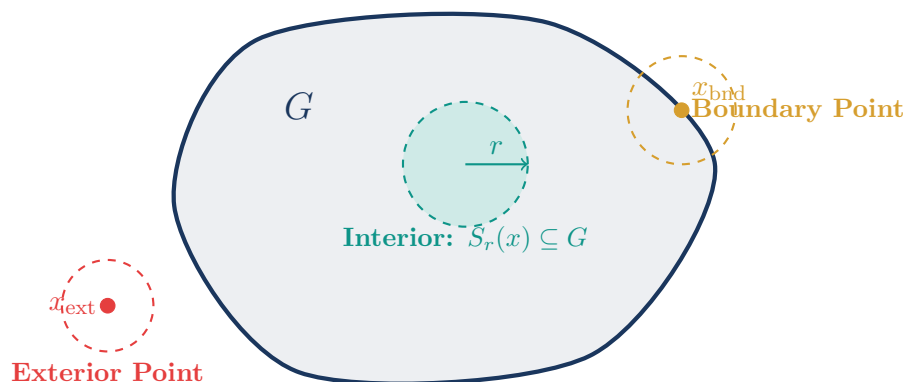
Definitions: Interior Point and Interior of a Set

Let G be a subset of a metric space (X, d) .

- A point $x \in G$ is called an **interior point** of G if and only if there exists an open sphere $S_r(x)$ with $r > 0$ such that:

$$S_r(x) \subseteq G \quad (16)$$

- The set of all interior points of G is called the **interior of G** , denoted by G° or $\text{Int}(G)$.
- Theorem:** A subset G is an **open set** if and only if every point of G is an interior point of G ($G = G^\circ$).



3 Module 3: Topology of Open Sets and Fundamental Theorems

3.1 Open Sets in a Metric Space

Definition: Open Set

A subset G of a metric space (X, d) is called an **open set** if and only if for each point $x \in G$, there exists a real number $r > 0$ such that:

$$S_r(x) \subseteq G \quad (17)$$

3.2 Fundamental Theorems on Open Sets with Complete Proofs

Theorem 1: The Empty Set $\emptyset$ and Full Space X are Open

Theorem: In any metric space (X, d) , the empty set $\emptyset$ and the full space X are open sets.

Proof:

1. **For the empty set $\emptyset$:** A set G is open if every $x \in G$ has an open ball contained in G . Since $\emptyset$ contains no elements, the condition holds vacuously. Thus, $\emptyset$ is open.
2. **For the full space X :** Let $x \in X$ be arbitrary. Choose any radius $r > 0$. By definition, the open sphere $S_r(x) = \{y \in X \mid d(y, x) < r\} \subseteq X$. Hence every point of X is an interior point of X , so X is open. $\square$

Theorem 2: Every Open Sphere is an Open Set

Theorem: In a metric space (X, d) , every open sphere $S_r(x_0)$ is an open set.

Proof:

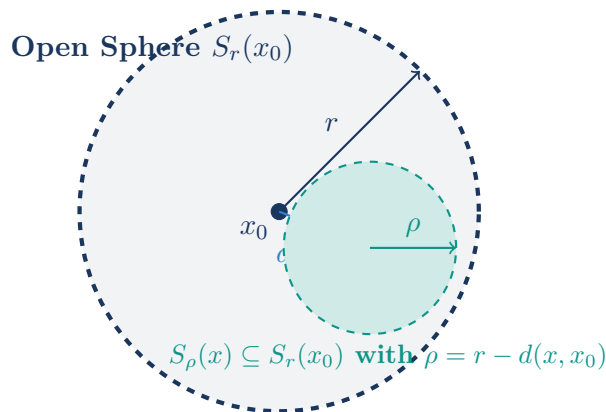
1. Let $S_r(x_0)$ be an open sphere centered at $x_0 \in X$ with radius $r > 0$.
2. Let x be an arbitrary point in $S_r(x_0)$. Then by definition:

$$d(x, x_0) < r \implies r - d(x, x_0) > 0 \quad (18)$$

3. Choose the radius $\rho = r - d(x, x_0) > 0$.
4. **Claim:** $S_\rho(x) \subseteq S_r(x_0)$.
5. To prove the claim, let $y \in S_\rho(x)$. Then $d(y, x) < \rho$.
6. By the triangle inequality [M4] on the metric d :

$$\begin{aligned} d(y, x_0) &\leq d(y, x) + d(x, x_0) \\ &< \rho + d(x, x_0) \\ &= (r - d(x, x_0)) + d(x, x_0) \\ &= r \end{aligned} \quad (19)$$

7. Since $d(y, x_0) < r$, it follows that $y \in S_r(x_0)$.
8. Therefore, $S_\rho(x) \subseteq S_r(x_0)$.
9. Since $x \in S_r(x_0)$ was arbitrary, every point in $S_r(x_0)$ is an interior point.
10. Hence, $S_r(x_0)$ is an **open set**. □



Theorem 3: Arbitrary Union of Open Sets is Open

Theorem: In a metric space (X, d) , the union of an arbitrary collection of open sets is an open set.

Proof:

1. Let $\{G_\lambda \mid \lambda \in \Lambda\}$ be an arbitrary collection of open sets in (X, d) , where Λ is an arbitrary index set.
2. Let $G = \bigcup_{\lambda \in \Lambda} G_\lambda$.
3. If $G = \emptyset$, then G is open by Theorem 1.
4. If $G \neq \emptyset$, let $x \in G$.
5. By definition of union, $x \in G_{\lambda_0}$ for some specific index $\lambda_0 \in \Lambda$.
6. Since G_{λ_0} is an open set, there exists a radius $r > 0$ such that:

$$S_r(x) \subseteq G_{\lambda_0} \quad (20)$$

7. Since $G_{\lambda_0} \subseteq \bigcup_{\lambda \in \Lambda} G_\lambda = G$, it follows that:

$$S_r(x) \subseteq G \quad (21)$$

8. Since $x \in G$ was chosen arbitrarily, every point of G is an interior point.
9. Hence, $G = \bigcup_{\lambda \in \Lambda} G_\lambda$ is an **open set**. □

Theorem 4: Finite Intersection of Open Sets is Open

Theorem: In a metric space (X, d) , the intersection of a finite number of open sets is an open set.

Proof:

1. Let $G_1, G_2, \dots, G_n$ be a finite collection of open sets in (X, d) .
2. Let $G = \bigcap_{i=1}^n G_i$.
3. If $G = \emptyset$, then G is open by Theorem 1.
4. If $G \neq \emptyset$, let $x \in G$.
5. By definition of intersection, $x \in G_i$ for every $i \in \{1, 2, \dots, n\}$.
6. Since each G_i is open, there exist positive real radii $r_1, r_2, \dots, r_n > 0$ such that:

$$S_{r_i}(x) \subseteq G_i, \quad \forall i \in \{1, 2, \dots, n\} \quad (22)$$

7. Define $r = \min\{r_1, r_2, \dots, r_n\}$. Since it is the minimum of a **finite** collection of strictly positive numbers, $r > 0$.
8. Since $r \leq r_i$ for each i , we have $S_r(x) \subseteq S_{r_i}(x) \subseteq G_i$ for all $i = 1, 2, \dots, n$.
9. Therefore:

$$S_r(x) \subseteq \bigcap_{i=1}^n G_i = G \quad (23)$$

10. Since $x \in G$ was arbitrary, $G = \bigcap_{i=1}^n G_i$ is an **open set**. $\square$

Important Counterexample: Infinite Intersection of Open Sets May NOT be Open

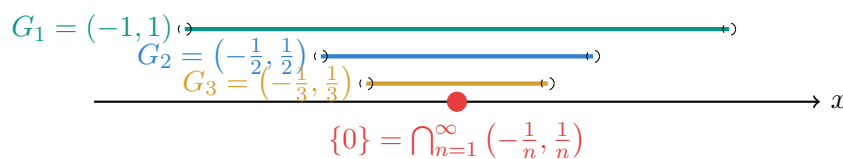
In $(\mathbb{R}, |\cdot|)$, consider the infinite family of open intervals:

$$G_n = \left(-\frac{1}{n}, \frac{1}{n}\right), \quad n \in \mathbb{N} \quad (24)$$

Each G_n is an open set in $\mathbb{R}$. However, their infinite intersection is:

$$\bigcap_{n=1}^{\infty} G_n = \bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\} \quad (25)$$

The singleton set $\{0\}$ is a **closed set** and is **not open** in $(\mathbb{R}, |\cdot|)$, because for any $r > 0$, the open ball $S_r(0) = (-r, r) \not\subseteq \{0\}$.



Infinite sequence of nested open intervals collapsing to closed singleton $\{0\}$

4 Module 4: Limit Points, Derived Sets, and Closed Sets

4.1 Limit Points (Accumulation Points)

Definition: Limit Point of a Set

Let (X, d) be a metric space and $A \subseteq X$. A point $p \in X$ is called a **limit point** (or **accumulation point** / cluster point) of A if and only if every neighborhood N_p of p contains at least one point of A different from p :

$$(N_p \setminus \{p\}) \cap A \neq \emptyset \quad (26)$$

Equivalently, for every $r > 0$:

$$(S_r(p) \setminus \{p\}) \cap A \neq \emptyset \quad (27)$$

4.2 Derived Sets and Examples

Definition: Derived Set A'

The set of all limit points of a subset A is called the **derived set** of A , denoted by A' or $d(A)$.

Solved Examples on Derived Sets

1. **Harmonic Sequence Set in $(\mathbb{R}, |\cdot|)$:** Let $A = \{1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots\}$. The only accumulation point is 0, because for every $\varepsilon > 0$, $(-\varepsilon, \varepsilon) \cap A$ contains infinitely many terms $\frac{1}{n}$. None of the elements $\frac{1}{n} \in A$ are limit points.

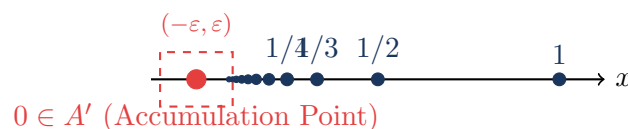
$$A' = \{0\}$$

2. **Open Interval in $(\mathbb{R}, |\cdot|)$:** Let $A = (0, 1)$. Every point in $(0, 1)$ as well as the endpoints 0 and 1 are limit points:

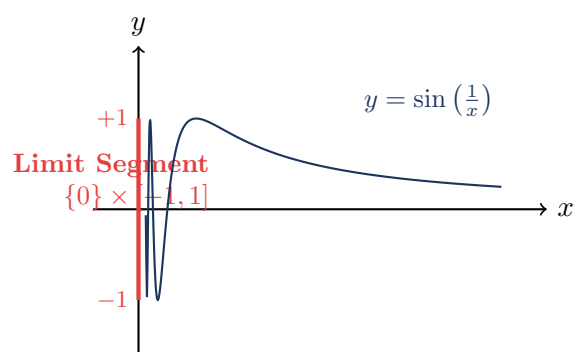
$$A' = [0, 1]$$

3. **Topologist's Sine Curve in $\mathbb{R}^2$:** Consider the subset $A = \{(x, y) \in \mathbb{R}^2 \mid y = \sin(\frac{1}{x}), x > 0\}$. As $x \rightarrow 0^+$, $\sin(1/x)$ oscillates between -1 and $+1$ infinitely often. The derived set A' contains the curve itself plus the entire vertical segment on the y -axis:

$$A' = A \cup (\{0\} \times [-1, 1])$$



Harmonic Set: Points accumulate at 0, so $A' = \{0\}$



Topologist's Sine Curve: Derived Set includes $\{0\} \times [-1, 1]$

4.3 Closed Sets in a Metric Space

Definition: Closed Set

Let (X, d) be a metric space. A subset $F \subseteq X$ is called a **closed set** if and only if its complement $F^c = X \setminus F$ is an open set in (X, d) .

Examples of Closed Sets

1. **Singleton Sets in $(\mathbb{R}, |\cdot|)$:** For any $x \in \mathbb{R}$, the singleton $\{x\}$ is closed because its complement:

$$\{x\}^c = (-\infty, x) \cup (x, \infty)$$

is the union of two open intervals (open sets), hence open. Thus, **every singleton set in a metric space is closed**.

2. **Closed Intervals in $(\mathbb{R}, |\cdot|)$:** $[a, b]$ is closed because $[a, b]^c = (-\infty, a) \cup (b, \infty)$ is open.

4.4 Topological Properties of Closed Sets

Duality Properties of Closed Sets (De Morgan's Laws)

1. The empty set $\emptyset$ and the whole space X are **closed sets** (since $\emptyset^c = X$ and $X^c = \emptyset$ are open).
2. The **intersection of an arbitrary collection** of closed sets is closed:

$$\left(\bigcap_{\lambda \in \Lambda} F_\lambda \right)^c = \bigcup_{\lambda \in \Lambda} F_\lambda^c \quad (\text{Union of open sets is open})$$

3. The **union of a finite number** of closed sets is closed:

$$\left(\bigcup_{i=1}^n F_i \right)^c = \bigcap_{i=1}^n F_i^c \quad (\text{Finite intersection of open sets is open})$$

4.5 Theorem: Limit Point Characterization of Closed Sets

Theorem: Closed Sets Contain All Their Limit Points

Theorem: In a metric space (X, d) , a subset $F \subseteq X$ is closed if and only if F contains all its limit points ($F' \subseteq F$).

Proof:

1. ($\implies$) **Necessary Condition:** Assume F is closed. We must show $F' \subseteq F$.
 - Let $p \in F^c = X \setminus F$.
 - Since F is closed, F^c is an open set.
 - By definition of an open set, there exists an open ball $S_r(p)$ such that $S_r(p) \subseteq F^c = X \setminus F$.
 - This implies $S_r(p) \cap F = \emptyset \implies (S_r(p) \setminus \{p\}) \cap F = \emptyset$.
 - Therefore, p is **not** a limit point of F ($p \notin F'$).
 - By contrapositive, if $p \in F'$, then $p \in F$. Thus $F' \subseteq F$.
2. ($\impliedby$) **Sufficient Condition:** Assume $F' \subseteq F$. We must show F is closed (i.e., F^c is open).
 - Let $x \in F^c = X \setminus F$.
 - Since $F' \subseteq F$, we have $x \notin F'$.
 - Since x is not a limit point of F , there exists an open sphere $S_r(x)$ such that:

$$(S_r(x) \setminus \{x\}) \cap F = \emptyset$$
 - Since $x \notin F$, we have $S_r(x) \cap F = \emptyset$, which implies $S_r(x) \subseteq X \setminus F = F^c$.
 - Since $x \in F^c$ was arbitrary, every point of F^c is an interior point of F^c .
 - Thus F^c is open, which proves that F is **closed**. □

5 Module 6: Master Quick Revision Cheat Sheet

1. Metric Axioms, Norms, and Standard Metrics

- **Metric Space Axioms** (X, d) : $\forall x, y, z \in X$:

1. **Non-negativity**: $d(x, y) \geq 0$.
2. **Identity of Indiscernibles**: $d(x, y) = 0 \iff x = y$.
3. **Symmetry**: $d(x, y) = d(y, x)$.
4. **Triangle Inequality**: $d(x, z) \leq d(x, y) + d(y, z)$.

- **Standard Metrics on $\mathbb{R}^n$** :

- Euclidean Metric (L_2): $d_2(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$
- Taxicab / Manhattan Metric (L_1): $d_1(x, y) = \sum_{i=1}^n |x_i - y_i|$
- Maximum / Chebyshev Metric (L_∞): $d_\infty(x, y) = \max_{1 \leq i \leq n} |x_i - y_i|$
- Discrete Metric on any set X : $d(x, y) = 0$ if $x = y$, and $d(x, y) = 1$ if $x \neq y$.
- Semi-metric (Pseudo-metric): Satisfies non-negativity, symmetry, and triangle inequality, but $d(x, y) = 0$ is possible for distinct points $x \neq y$ (e.g., $d(x, y) = |x^2 - y^2|$ on $\mathbb{R}$).

- **Matrix Frobenius Norm on $M_{m \times n}(\mathbb{R})$** :

$$d(A, B) = \|A - B\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n (a_{ij} - b_{ij})^2} = \sqrt{\text{Tr}((A - B)^T (A - B))}$$

2. Open Sets, Spheres, and Neighborhoods

- **Open Sphere (Ball)**: $S_r(x_0) = B(x_0, r) = \{x \in X \mid d(x, x_0) < r\}$.
- **Closed Sphere (Ball)**: $\bar{S}_r(x_0) = \{x \in X \mid d(x, x_0) \leq r\}$.
- **Interior Point**: $x \in A$ is an interior point of A if $\exists r > 0$ such that $S_r(x) \subseteq A$.
- **Open Set**: $A \subseteq X$ is open if every point of A is an interior point of A (i.e., $A = A^\circ$).
- **Fundamental Theorem on Open Balls**: Every open sphere $S_r(x_0)$ in a metric space is an **open set**.
 - Proof radius choice: For any $x \in S_r(x_0)$, choose radius $\rho = r - d(x, x_0) > 0$, then $S_\rho(x) \subseteq S_r(x_0)$.
- **Topological Properties of Open Sets**:
 - $\emptyset$ and X are open sets.
 - The **arbitrary union** of open sets is open: $\bigcup_{\lambda \in \Lambda} U_\lambda$ is open.
 - The **finite intersection** of open sets is open: $\bigcap_{i=1}^n U_i$ is open.
 - Infinite intersections of open sets need not be open (Counterexample: $\bigcap_{n=1}^\infty (-\frac{1}{n}, \frac{1}{n}) = \{0\}$, which is not open in $\mathbb{R}$).

3. Limit Points, Derived Sets, and Closed Sets

- **Limit Point (Accumulation Point):** A point $p \in X$ is a limit point of $A \subseteq X$ if every open sphere centered at p contains at least one point of A distinct from p :

$$\forall r > 0, \quad (S_r(p) \setminus \{p\}) \cap A \neq \emptyset$$

(Note: p need not belong to A).

- **Derived Set A' :** The set of all limit points of A .
 - For $A = (a, b) \implies A' = [a, b]$.
 - For Harmonic Set $A = \{1/n \mid n \in \mathbb{N}\} \implies A' = \{0\}$.
 - For $A = \mathbb{Q} \implies A' = \mathbb{R}$.
 - For $A = \mathbb{Z} \implies A' = \emptyset$.
- **Isolated Point:** A point $x \in A$ which is **not** a limit point of A ($\exists r > 0$ such that $S_r(x) \cap A = \{x\}$).
- **Closed Set:** A subset $F \subseteq X$ is closed if its complement $F^c = X \setminus F$ is an open set.
- **Theorem (Limit Point Characterization):** $F \subseteq X$ is closed if and only if F contains all its limit points:

$$F \text{ is closed} \iff F' \subseteq F$$

- **Closure of a Set $\bar{A}$:** $\bar{A} = A \cup A'$. $\bar{A}$ is the smallest closed set containing A .
- **Topological Properties of Closed Sets:**
 - $\emptyset$ and X are closed sets.
 - The **arbitrary intersection** of closed sets is closed: $\bigcap_{\lambda \in \Lambda} F_\lambda$ is closed.
 - The **finite union** of closed sets is closed: $\bigcup_{i=1}^n F_i$ is closed.